

Winkler Oxygen method for seawater analysis

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Chemicals

Recipes can be founded in [Dickson \(1996\)](#), [IAP \(2006\)](#), and [Langdon C. \(2010\)](#).

- Manganese II Chloride - MnCl_2 (3M)
- Sodium hydroxide/ sodium iodide - NaOH (8M)/ NaI (4M) ****store in opaque bottle**
- Sulphuric acid - H_2SO_4 (5M)
- Potassium KIO_3 (0.01N = 0.00167 M)
- Sodium Thiosulphate - $\text{Na}_2\text{S}_2\text{O}_3$ (stock solution of **0.18M**, working solution in **0.018M**)

Equipment

For sampling

- Reagents bottles (MnCl_2 , NaOH/NaI, H_2SO_4) with attached 1ml dispenser cap
- A carrying stand to hold all reagent bottles.
- Biological Oxygen Demand (BOD) bottles with matching lid number (volume calibrated)
- A sampling storage box with opaque cover (i.e. a piece of wood or black garbage bag)
- Tygon tube (double to triple the length of the BOD bottles)

For laboratory analysis

- Sensor probe for air temperature and pressure
- Winkler system
- Magnetic stir bar
- BOD samples
- 2 empty BOD bottles (1 for blanks, 1 for standard)
- 1ml pipette and 10ml repeating dispenser for KIO_3 (Must calibrate for accuracy!)

Others

- 100ml graduating cylinder (for dilution)
- A plastic bucket with close lid for acid bath
- Washing bottles for milli-q water
- Waste beaker and 10L jug for chemical disposal.
- Secure spots in 4 deg C fridge

Sampling procedure (for CTD rosette)

Pre-sampling check

- ❑ **Pump the reagent bottles for a few times into a waste bottle.** Get rid of air bubbles in the dispensing tube that could affect the accuracy of the dispensing volume.
- ❑ **Make sure all BOD bottles have matching lids.** The volume of BOD bottles is pre-measured for oxygen calculation, using wrong lid will lead to inaccurate results.

Sampling Steps

1. Connect the valve of Niskin bottle with a Tygon tube.
2. Insert the other end of the tube into the BOD bottle upside down.
3. Turn on the valve slowly until getting a decent flow.
4. Rinse the bottle upside down and the lid for a few second.
5. Turn over and start filling the BOD bottle.
Important: The tube must insert all the way to the bottom of the bottle.
6. Overflow the bottle for 2 – 3 volumes.
Technique: Count the time needed to fill up the bottle, then wait for double of the time. If it takes 4s to fully fill up, then wait for another 4-8s.
7. Slowly take out the Tygon tube and turn off the Niskin valve when the tube is out.
Important: While pulling out the Tygon tube, use the tube to remove any gas bubble around the bottle rim.
8. Immediately add the reagents (1st : 1 ml MnCl_2 , 2nd : 1 ml NaOH/NaI) into the sample.
Important: Insert the tip of the dispenser into the surface of the sample to reduce contamination of oxygen in the air.
9. Put the cap on and start shaking the bottle for 30 sec – 1 min.
10. Once complete mixing, seal the gap between the bottle neck and the lid with milli-q water.
11. Place the samples in the storage box and cover with the dark lid.

Post-sampling procedure

- **20 mins after sampling:** shake the bottle again for 30s and seal the bottle gap with milli-q water.
- Store the samples in 4 deg C fridge.
- Analyse 2 – 4 hours later after all precipitates are settled at the bottom.
- Continue to store in 4 deg C fridge overnight if won't be analysed on the same day.

Winkler measurement at sea



Figure 1. An example of Winkler system set up, taken from CE22009 cruise onboard R/V Celtic Explorer.

Equipment list

- A sensor probe anchored next to the system (measures (air temperature & pressure)
- Winkler system
- Magnetic stir bar
- Reagents
- Samples
- Milli-q water
- Waste bottle.

Daily Routine

1. Preparing the system before analysis

Check list:

- ☐ Turn on temperature probe.
- ☐ Swirl the reagent bottle ($\text{Na}_2\text{S}_2\text{O}_3$).
- ☐ Turn the vent on top of the $\text{Na}_2\text{S}_2\text{O}_3$ bottle to the “open” position.
- ☐ Flush out old reagent and get rid of air bubbles in the tube.
- ☐ Place electrode and reagent line on the stirring stand.
- ☐ Check the level of all reagents and make sure it’s not too low.

Before starting sample analysis:

- **Reagent blanks:** Typically at the beginning of the day before any sample is analysed, or if any reagent is changed to a new batch.
- **Standard:** before each batch of samples

Examples	Analysis sequence
Analyse 2 batch of samples (CTD1, CTD2) + No reagent change	Blanks → Standard → CTD1 → Standard → CTD2
Analyse 2 batch of samples (CTD1, CTD2) + Reagent change after CTD1	Blanks → Standard → CTD1 → Blanks → Standard → CTD2

Table 1. Examples of analysis sequence.

2. Preparing blanks and standard

1. Swirl the reagent bottle with 0.01N KIO₃.
2. Pour some KIO₃ into a clean beaker (~ 50 - 70 ml)
3. Pour some milli-q water into a clean BOD bottle.
4. Dispensing KIO₃ into the BOD bottle
 - a. **For blanks:** use a 1 ml pipette to dispense the appropriate amount.
 - b. **For standard:** use a 10ml repeating dispenser (with 50 ml syringe volume)
 - i. Draw KIO₃ from the beaker.
 - ii. Dispense 3 times (~30ml) back into the beaker to remove air bubbles in the dispenser.
 - iii. Dispense 10ml KIO₃ into the BOD bottle.
5. Add 1ml of H₂SO₄
6. Add 1ml of NaOH/NaI
7. Add 1ml of MnCl₂
8. Top up with milli-q water and leave 1 – 2 inches space below bottle mouth.

	KIO ₃ (0.01N)	H ₂ SO ₄	NaOH/NaI	MnCl ₂	Milli-q
Blank 1	1 ml	1 ml	1 ml	1 ml	Top up until below bottle neck
Blank 2	2 ml	1 ml	1 ml	1 ml	
Blank 3	4 ml	1 ml	1 ml	1 ml	
Blank 4	8 ml	1 ml	1 ml	1 ml	
Standard	10 ml	1 ml	1 ml	1 ml	



Table 2. Reagent volumes and adding sequences (black arrow) for blanks and standard.

3. Running samples

*** Make sure DO samples are warm up to room temperature***

1. Open the bottom cap gently.
2. Wash the magnetic stirrer with milli-q water and put into the BOD bottle slowly.
3. Dispense 1m H₂SO₄ acid into the sample.
4. Put the sample on the stirring stand.
5. Place the electrode and reagent tube into the BOD bottle, and make sure the sample doesn't overflow (water level should be below the level of bottle neck).

4. Shutting down at the end of day

- Rinse the electrode and reagent tube with milli-q water
- Place the electrode back into the storage tube (it should fill with storage solution or milli-q water)
- Turn the vent on top of the $\text{Na}_2\text{S}_2\text{O}_3$ reagent bottle to the “close” position (avoid evaporation).
- All samples should be disposed properly in a waste container. Ideally, a 10L plastic jug should be used for waste disposal in the lab, with a written list of chemical on the jug.

Bottle Storage and Maintenance

- Keep the BOD bottles dry and safely secure in a box with bottle slots.
- BOD lid could be jammed in the bottle while dry. A trick to avoid this is to **put a small piece of paper between the lid and the bottle**. Be careful when you remove the lid from the bottle and make sure the paper doesn't fall into the bottle.
- Oxide (black precipitates) could form near the bottle rim and lids after a few uses, and they can cause contamination to the DO samples. To remove the oxide, **soak the bottles and lid in 10% HCl acid bath** for a few hours, until no precipitate left on the bottles (the acid bath should turn slightly yellow). Rinse the bottles with milli-q water bath and leave them in a tray to dry.



Figure 2. An example of oxide formation (black stains) on the BOD bottle.

Reference

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